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Wavelenght calibration with WR lines

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Abstract

Key words: stars: evolution — stars: supergiants — stars: massive

1 Introduction

Wolf–Rayet stars exhibit strong emission lines in their spectra. They can be identified even at the low resolution of the Gaia BP/RP spectra. An example of line identification is provided in Marin et al. (2024).

A catalog of Wolf-Rayet stars in the Milky Way is mantained by Prof. Crowther and is available on Crowther's webpage.

2 A set of reference spectra

The database compiled by Crowther comprises about 700 Wolf–Rayet (WR) stars, all of which are already identified with Gaia DR3 sources. For the present study, I have used a subset of 397 stars, assembled from the lists of van der Hucht (2001), Mauerhan et al. (2011), Shara et al. (2012), and Kanarek et al. (2015).

These 397 stars were cross-correlated with the 2MASS database and subsequently with the Gaia DR3 catalog using the 2MASS identifiers, yielding 204 secure sources with available BP/RP spectra. The Gaia sources were verified against the SIMBAD database.

Known binary stars were excluded, as well as stars with poor BP/RP spectra or spectra with ambiguous classifications (reported as WN or WC but showing mixed features indicative of binarity). Only Gaia point sources with RUWE values smaller than 1.4 were retained.

In the end, a list of 82 WN stars and 31 WC stars was obtained.

As noted above, additional stars can be incorporated from Crowther's comprehensive compilation (but, this will require a new round of visual inspection, cannot be done automatically).

3 High-resolution spectra

I have quickly found 16 nice spectra.

UVES_POP library contains the spectra of 8 WR stars. The official UVES pipeline (as documented in the ESO UVES Pipeline User Manual) uses air wavelengths for the reference line tables in wavelength calibration (e.g., ThAr arc lamps).

Aadland et al. (2022) analyzed high-quality UV, optical, and near-IR (NIR) data on six WC-type stars and two WO. The spectra are on VIZIER. A ThAr lamp was used to make wavelength calibrations and data reduction was done using the IDL FIRE pipeline (Simcoe et al. 2013)

A direct comparison of BPRP spectra and high-resolution is therefore possible.

4 News of PNs

In preparation for the calibration of Euclid data, several planetary nebulae (PNs) were observed with VLT/X-shooter (Euclid Collaboration et al. 2023). The corresponding FITS files are available at VizieR (Fits at VIZIER).

section References

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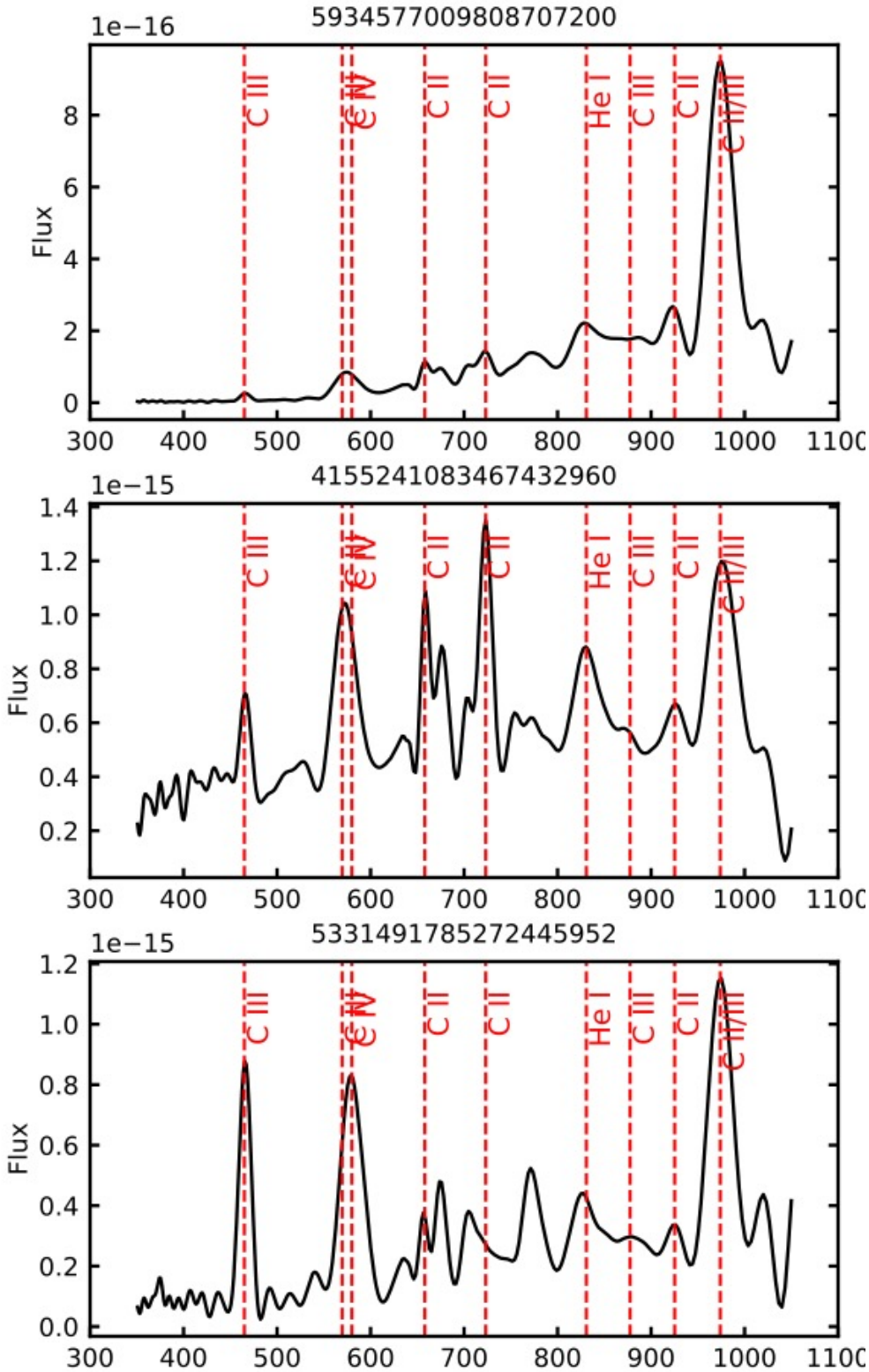

Fig. 1. Example of a WC spectrum: a WC9 at the top, a WC9d in the middle, and a WC6 in the lower panel. Lines prominent in WC spectra are plotted in red, while those characteristic of WN stars are shown in green. Note that the lines around 550nm vary with spectral type; therefore, precise knowledge of the spectral type is required, as indicated in the table.

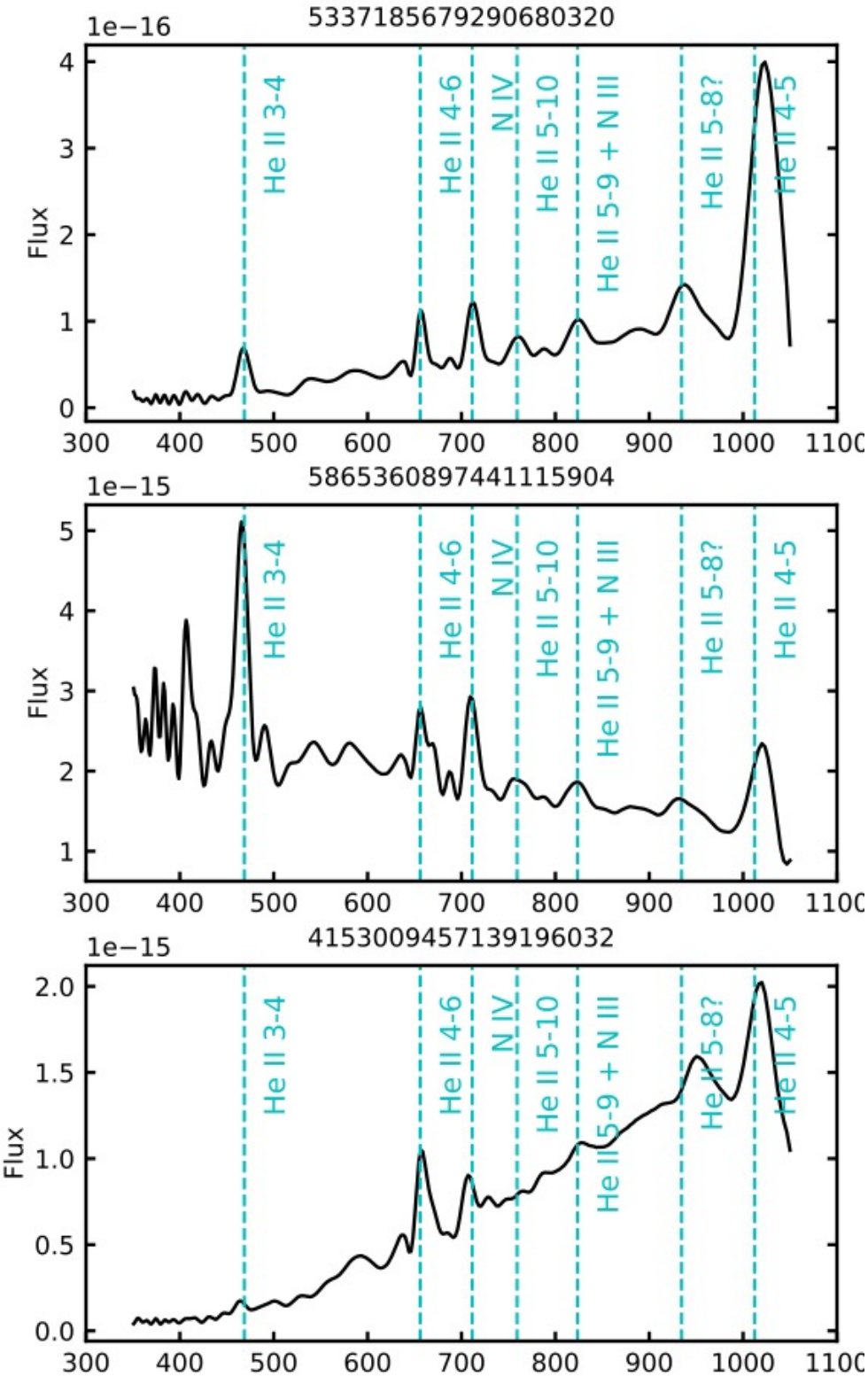

Fig. 2. Example of WN spectra: a WN4-s at the top, a WN7 in the middle, and a WN8h in the bottom panel. Lines prominent in WC spectra are plotted in red, while those characteristic of WN stars are shown in green.

Table 1. Prominent lines in WC stars (red lines)

Line Å	Element	reference	Atom.L.L. air	Atom.L.L. vacuum	types
4650	CIII[a]	Marin et al. (2024)	4647-4652	4648-4653	all WC
5696	CIII	Marin et al. (2024)	5695.916	5697.496	WC8/WC9
5800	CIV	Crowther (2007)	5801.35	5802.95	WC5,WC6,WC7
6580	CII	Marin et al. (2024)	6578-6582	6579-6584	
7231.36	CII	Marin et al. (2024)			
8308	He I	He I $2^1S - 2^1P$ (singlet)			WC8/WC9
8772-8780	CIII	Marin et al. (2024)			
9238-9267	CII	Marin et al. (2024)			WC8/WC9
9733-9747	CII/III blend	Marin et al. (2024)			all WC

- Lines are taken from Marin et al. (2024), Aadland et al. (2022), and Crowther (2007).
- (a)= This is the well-known C III multiplet (components typically listed as 4647.42 Å, 4650.25 Å, 4651.47 Å in vacuum; the "4650" is shorthand for the central/strongest component or the whole blend at 4650 Å).

Table 2. Prominent lines in WR stars (green lines)

Line (air) Å	Element	Reference	AtomicLL air	AtomicLL vacuum	Types
4686	He II (3-4)	1	4685.71	4687.02	
6560	He II (4-6)	1	6560.10	6561.91	
7115	NIV	1	7109.4-7123.0	7111.3-7124.9	
7592	He II (5-10)	2,3	7592.754	7594.844	
8237	He II (5-9)+N III	2,4	8224-8239	8226.9-8241.5	
9344.9	He II (5-8)?	5	9344.94	9347.50	WN3, WN4, WN5, WN6
10124	He II (4-5)	6,7	10123.61	10126.39	
10830	He I	6,7,8	10829-10830	10832-10833	

- Lines are taken from 1=Marin et al. (2024), 2=Hamann et al. (1995), 3==Hamann et al. (2006), 4=Nugis et al. (2008), 5=Peter van Hoof - Atomic line list, 6=Howarth & Schmutz (1992), 7=Vreux et al. (1990), 8=Groh et al. (2007).

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paolo reference for the PN is Paterson 2023 A&A 674, A172

Table 3. The spectra of the UVES_POP

UVES_id Comment	source_id	WRtype	WRsub
HD050896	2922367976673391232	WN	WN4b
HD093131	5254268071479968512	WN	WN6ha-w
HD094910 poor quality	5338220285385672064	WN	WN11h
HD151932	5968019618048049664	WN	WN7h
HD092740	5350370026290390912	WN	WN7h+O9III-V
HD156385	5952743072247166464	WC	WC7
HD113904	5858915766471945984	WC	WC5+B0III
HD152270	5966509541883497984	WC	WC7+O5-8

Table 4. UVES_POP: Mean ± standard deviation of Lorentzian fit parameters for WN spectral lines: rest wavelength λ_0, mean centroid $\langle\lambda_c\rangle$, equivalent width $\langle EW\rangle$, half-width at half-maximum $\langle\gamma\rangle$, and number of valid fits N.

Line	λ_0 [nm]	$\langle\lambda_c\rangle$ [nm]	$\langle EW\rangle$ [nm]	$\langle\gamma\rangle$ [nm]	N
He II 3-4	468.600	468.6525 ± 0.0896	8.0491 ± 5.2120	1.0000 ± 0.0000	4
He II 4-6	656.000	656.4125 ± 0.1081	5.7504 ± 1.4243	1.0000 ± 0.0000	4
N IV	711.500	711.9450 ± 0.5004	2.4537 ± 1.7528	1.0000 ± 0.0000	4
He II 5-10	759.270	759.5100 ± 0.1175	1.2575 ± 1.0372	1.0000 ± 0.0000	4
He II 5-9 + N III	823.700	823.8550 ± 0.0866	1.3886 ± 0.9808	1.0000 ± 0.0000	4
He II 5-8?	934.490	934.4900 ± 0.0000	3.2873 ± 3.0267	0.8000 ± 0.0000	4
He II 4-5	1012.400	1012.4000 ± 0.0000	16.5668 ± 7.5562	0.8000 ± 0.0000	4

1 WR in UVES_POP

The spectra of the UVES_POP are already at rest wavelength. WN stars have strong He II lines, which are nicely symmetric and peaked. WN stars may yield a solid wavelength calibration.

WC stars are characterise by CII and CII lines, mostly with flat profile.

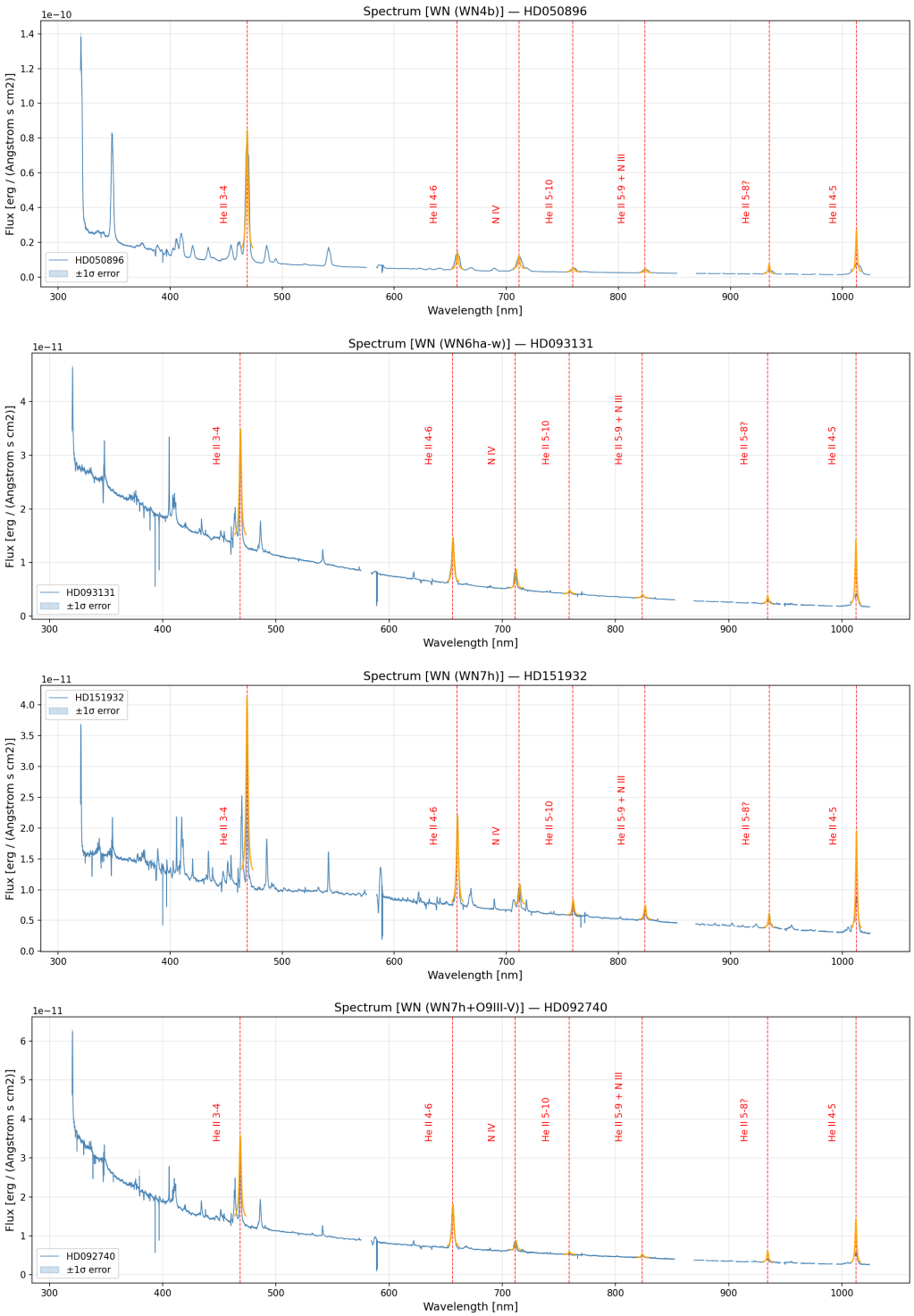

Fig. 3. High-resolution spectra of WR WN-type stars from the UVES_POP library.

Table 5. Summary of radial velocity measurements for WN stars from UVES_POP.

Line	$\Delta\lambda$ [nm]	$v_{\rm rad}$ [km/s]
He II 3-4 (469)	+0.0525	$+34 \pm 57$
He II 4-6 (656)	+0.4125	$+189 \pm 49$
N IV (712)	+0.4450	$+188 \pm 211$
He II 5-10 (759)	+0.2400	$+95 \pm 46$
He II 5-9+N III (824)	+0.1550	$+56 \pm 31$
Weighted mean		$+107 \pm 22$

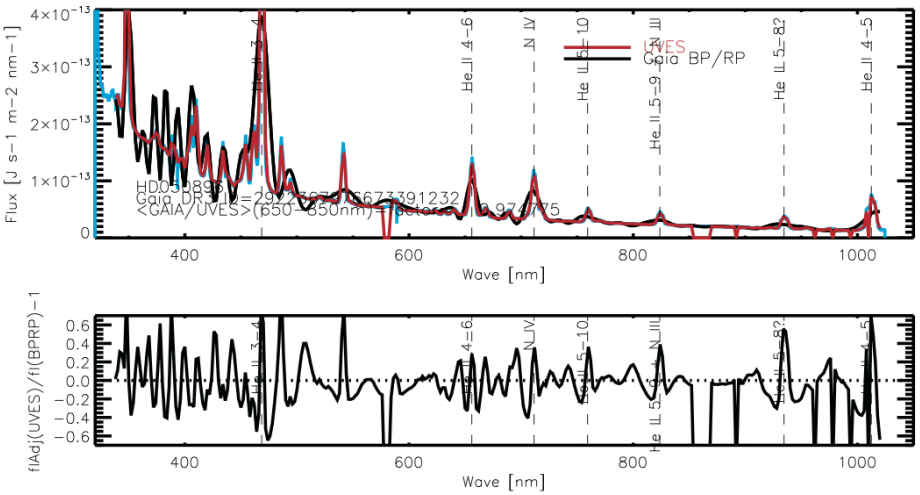

Fig. 4. BPRP spectra of WR WN-type stars from the UVES_POP library:2922367976673391232

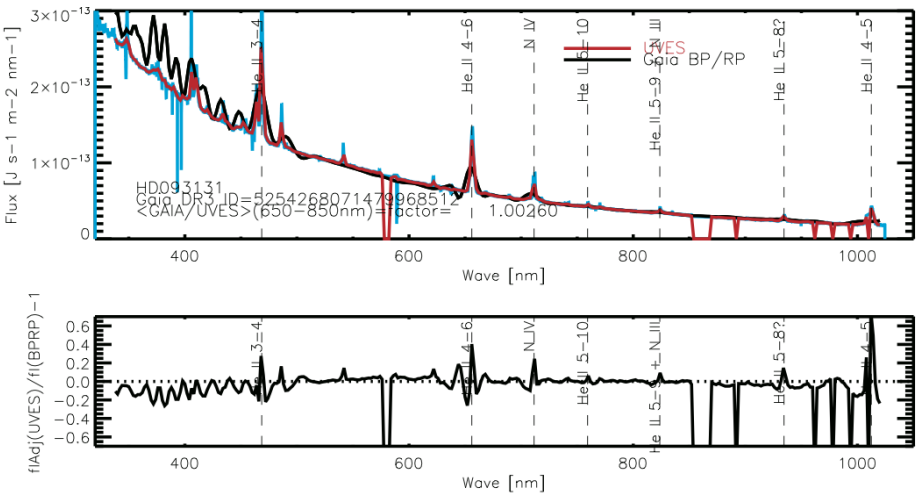

Fig. 5. BPRP spectra of WR WN-type stars from the UVES_POP library:5254268071479968512

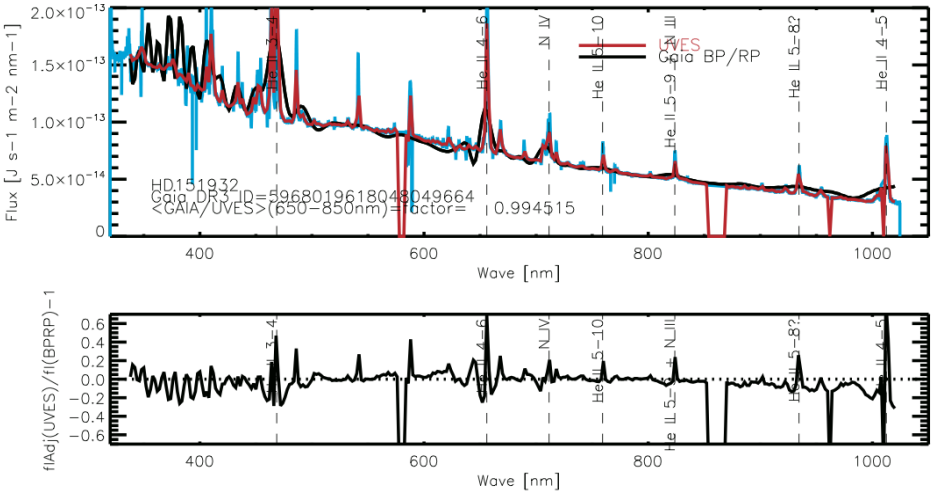

Fig. 6. BPRP spectra of WR WN-type stars from the UVES_POP library: 5968019618048049664

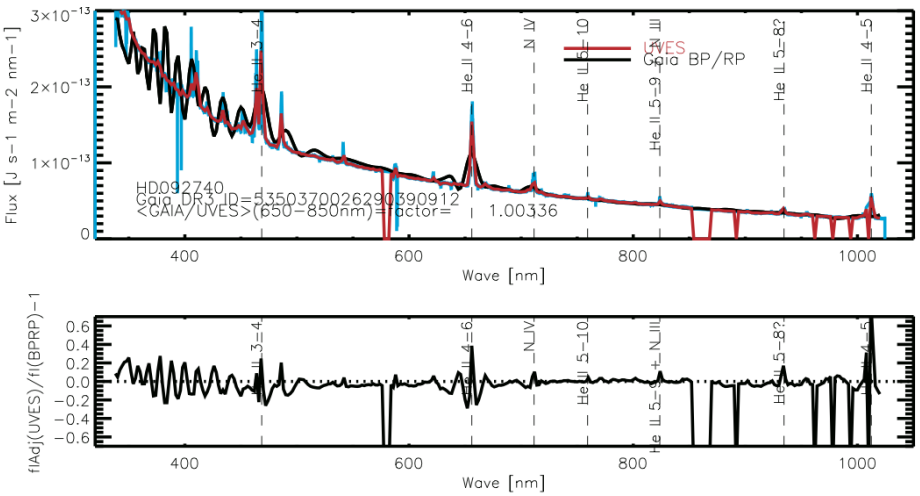

Fig. 7. BPRP spectra of WR WN-type stars from the UVES_POP library: 5350370026290390912

Table 6. Gaussian fit centroids λ_c [nm] per WC star (UVES_POP). Dashes indicate failed fits.

Star	C III	C III	C II	C II	He I	C III	C II	C II/III
HD156385	465.8128	...	657.0786	723.1360	830.8000	877.6000	925.2500	974.0000
HD113904	465.4617	...	656.4759	723.1360	830.8000	877.6000	925.2522	974.0000
HD152270	466.0429	...	657.4349	723.1360	830.8000	877.6000	925.2508	974.0000

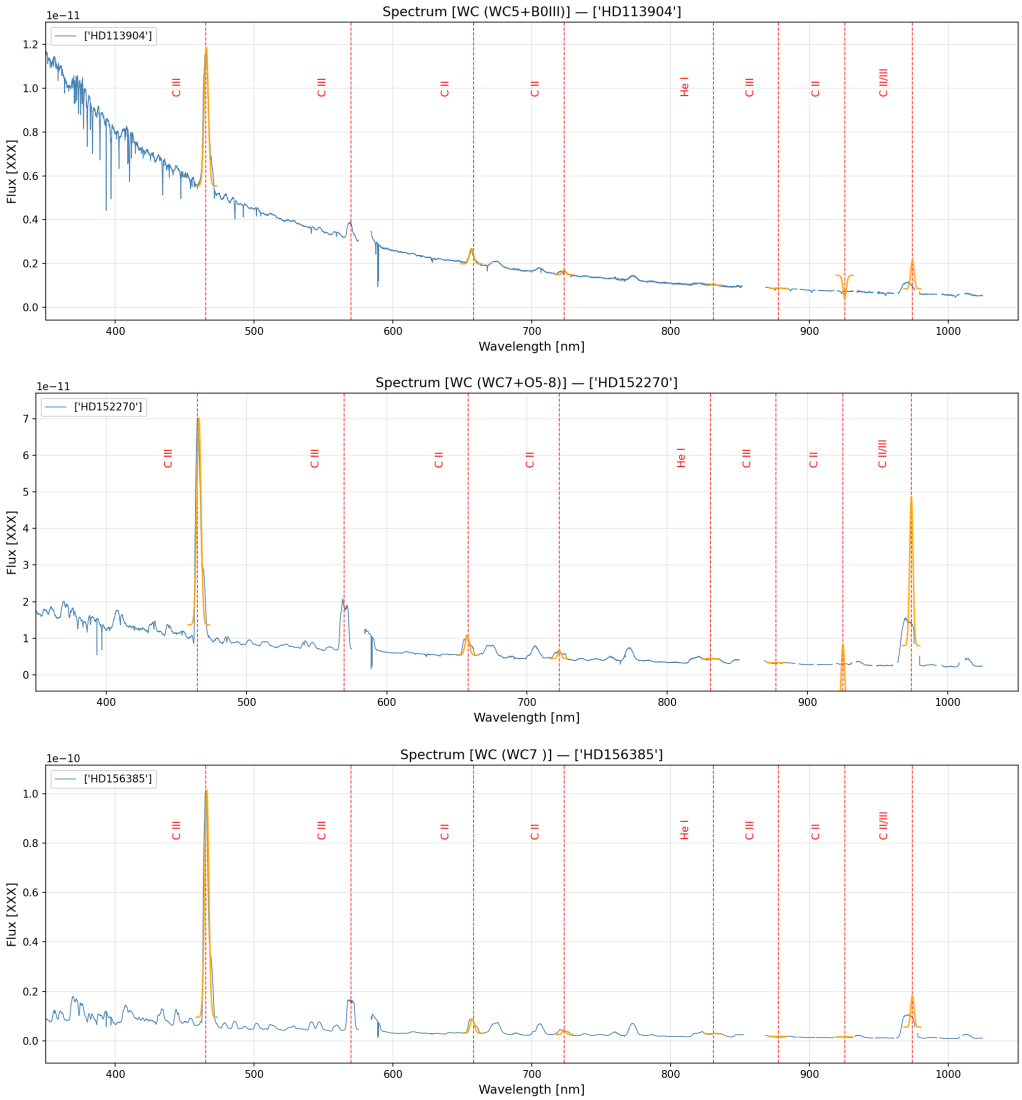

Fig. 8. These are the three spectra of WC stars available in the UVES_POP library. The lines are broad and not well picked. Furthermore, the CII/III blend reported at 9733-9747 Å by Marin et al. (2024) appears a bit offset. It is definitely better to use only WN stars.

Table 7. Mean ± standard deviation of Gaussian fit parameters for WC spectral lines (UVES_POP): rest wavelength λ_0, mean centroid $\langle\lambda_c\rangle$, equivalent width $\langle EW\rangle$, half-width at half-maximum $\langle\gamma\rangle$, and number of valid fits N.

Line	λ_0 [nm]	$\langle\lambda_c\rangle$ [nm]	$\langle EW\rangle$ [nm]	$\langle\gamma\rangle$ [nm]	N
C III	465.000	465.7725 ± 0.2927	18.7332 ± 16.2481	1.5000 ± 0.0000	3
C III	569.600	...	...	...	0
C II	658.000	656.9964 ± 0.4848	4.0442 ± 2.8401	1.5000 ± 0.0000	3
C II	723.136	723.1360 ± 0.0000	1.3917 ± 0.9258	1.2000 ± 0.0000	3
He I	830.800	830.8000 ± 0.0000	0.1973 ± 0.0508	1.2000 ± 0.0000	3
C III	877.600	877.6000 ± 0.0000	0.0762 ± 0.0657	1.2000 ± 0.0000	3
C II	925.250	925.2510 ± 0.0011	-2.4841 ± 2.8353	1.2000 ± 0.0000	3
C II/III	974.000	974.0000 ± 0.0000	8.9676 ± 5.6576	1.2000 ± 0.0000	3

Table 8. Best line in the WC sample from UVES_POP.

Line	$\Delta\lambda$ [nm]	$v_{\rm rad}$ [km/s]
C III 465	+0.7725	$+498 \pm 189$

2 AAdland's sample

Six WC were observed by Aadland et al. (2022). The spectra cover the entire range of the BPRP spectra. The targets belong to the LMC, which has a systemic heliocentric radial velocity of $-262\ \mathrm{km\,s^{-1}}$.

The paper does not state wavelength correction to bring it at rest wavelength.

Both lines, C III 465 and C IV 580, are consistently redshifted after the 250 km/s LMC correction is applied. The residual +620 km/s is larger than the resolution element velocity (c/R = 3000 km/s) divided by 5, so it's marginal at R=100.

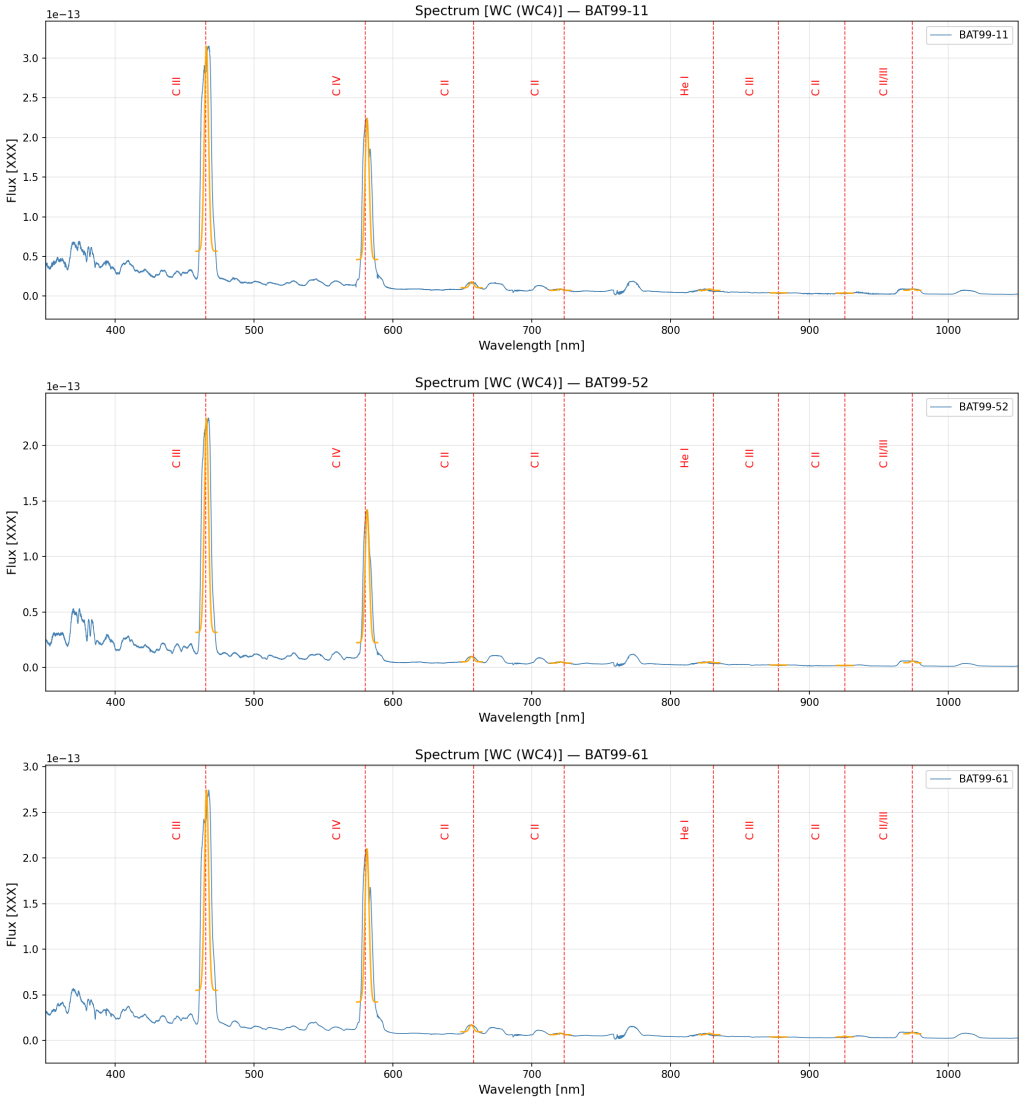

Fig. 9. WC stars by Aadland et al. (2022). A correction of -262 km s^{-1} is applied to all spectra (LMC).

Table 9. Gaussian fit centroids λ_c [nm] per WC star by Aadland et al. (2022). A correction of -262 km s^{-1} is applied to all spectra (LMC). Dashes indicate failed fits.

Star	C III	C IV	C II	C II	He I	C III	C II	C II/III
BAT99-11	465.5792	581.2948	656.5873	720.3305	827.8000	877.6000	925.2500	974.0000
BAT99-52	465.5671	581.3734	656.6485	720.5117	827.8000	877.6000	925.2500	974.0000
BAT99-61	465.6694	581.2508	656.4277	720.3693	827.8000	877.6000	925.2500	974.0000
BAT99-8	465.5077	581.2416	656.4565	720.1571	827.8000	877.6000	925.2500	974.0000
BAT99-9	465.9042	581.1873	656.4275	720.1360	827.8000	877.6000	925.2500	974.0000
BAT99-90	465.6477	581.4135	656.5531	720.7831	827.8000	877.6000	925.2500	974.0000

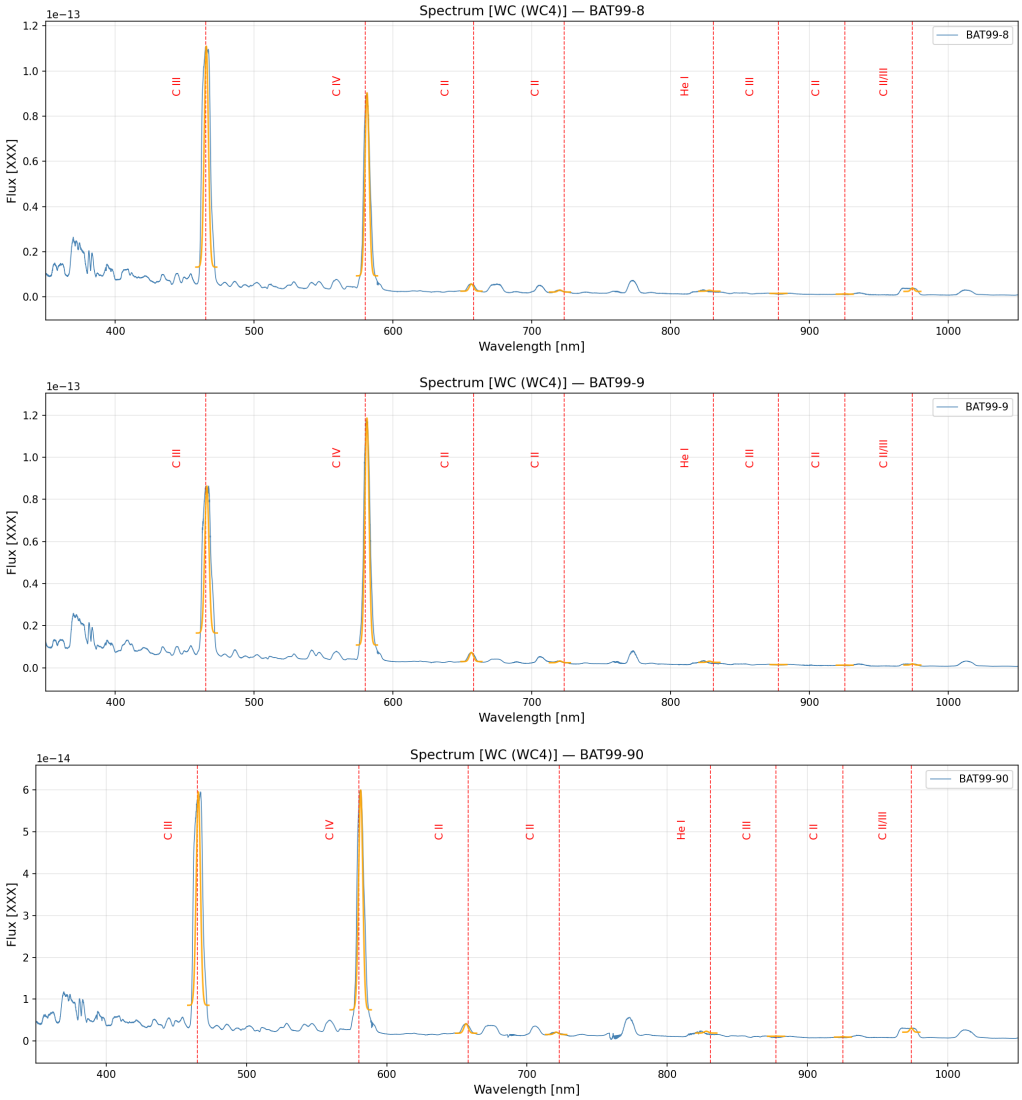

Fig. 10. WC stars by Aadland et al. (2022). The first two lines are fitted using the basis of the line (not the peak). A correction of $-262\ \mathrm{km\,s^{-1}}$ is applied to all spectra (LMC).

Table 10. Mean ± standard deviation of Gaussian fit parameters for WC spectral lines by Aadland et al. (2022): rest wavelength λ_0, mean centroid $\langle\lambda_c\rangle$, equivalent width $\langle EW\rangle$, half-width at half-maximum $\langle\gamma\rangle$, and number of valid fits N. A correction of -262 km s^{-1} is applied to all spectra (LMC).

Line	λ_0 [nm]	$\langle\lambda_c\rangle$ [nm]	$\langle EW\rangle$ [nm]	$\langle\gamma\rangle$ [nm]	N
C III	465.000	465.6459 ± 0.1393	20.2205 ± 4.9794	1.5000 ± 0.0000	6
C IV	580.000	581.2936 ± 0.0855	24.3711 ± 9.4683	1.5000 ± 0.0000	6
C II	658.000	656.5168 ± 0.0929	4.0203 ± 1.0900	1.5000 ± 0.0000	6
C II	723.136	720.3813 ± 0.2415	1.2043 ± 0.2405	1.5000 ± 0.0000	6
He I	830.800	827.8000 ± 0.0000	0.8239 ± 0.1147	1.5000 ± 0.0000	6
C III	877.600	877.6000 ± 0.0000	0.0927 ± 0.0439	1.2000 ± 0.0000	6
C II	925.250	925.2500 ± 0.0000	0.3301 ± 0.1307	1.2000 ± 0.0000	6
C II/III	974.000	974.0000 ± 0.0000	1.1914 ± 0.4088	1.2000 ± 0.0000	6

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he centroid could be determined only for strong lines. For the faintest the rest wavelength is assumed, and the errors are set to null.

Table 11. Best two lines in the WC sample of Aadland et al. (2022).

Line	$\Delta\lambda$ [nm]	vel_rad [km/s]
C III 465	+0.646	$+416 \pm 90$
C IV 580	+1.294	$+669 \pm 44$
Weighted mean	+1.167	$+620 \pm 40$